Spatial–Spectral Oriented Triple Attention Network for Hyperspectral Image Denoising

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***Abstract*— Hyperspectral images (HSIs) often suffer from degradation caused by mixed noise, leading to a decline in the performance of subsequent advanced applications. To eliminate noise and improve image quality, transformer-based approaches have been successfully employed. Nevertheless, these strategies often involve large-scale modeling and tedious layer normalization, which causes inefficiencies during the denoising process. Additionally, the neglect of local spectral correlations in HSIs damages the physical properties in recovery, resulting in poor generalization and inefficient denoising performance. To address these problems, we propose an efficient spatial–spectral oriented triple attention network, dubbed S^{2}OTAN, for HSI denoising. Specifically, to fully exploit the physical properties of HSIs, we impose spatial and spectral multiscale hybrid attention in the single-transformer block side-by-side to fuse spatial–spectral information in a parallel manner. For spatial feature extraction, we introduce hybrid spatial attention by constructing attention maps for pixels within and across windows to exploit the local and global similarity in spatial and improve computational efficiency. For spectral feature exploration, we utilize spectral partitioning operations to enhance the adjacent spectral dependences of HSIs and capture contextual information related to correlations. Consequently, our method exhibits a robust feature representation capability for removing mixed noise in HSIs. Extensive experiments on synthetic and real-world noisy scenarios demonstrate that the proposed approach outperforms other state-of-the-art approaches among quantitative metrics and visual effects. For the sake of reproducibility, the code is available at: https://github.com/Zilong-Xiao/S2OTAN.**

***Index Terms*— Computational efficiency, hyperspectral images (HSIs), mixed noise, nonlocal similarity, triple attention network.**

I. Introduction

HYPERSPECTRAL imagers can acquire spatial structural information and abundant spectral fingerprint data. Benefit from this characteristic, hyperspectral imagers are generally applied in the fields of object detection [1], [2], [3], industrial monitoring [4], agriculture [5], and medical diagnosis [6]. Nevertheless, during the acquisition, hyperspectral images (HSIs) are often subjected to disturbances caused by mixed noise and degradation because of limitations in imaging conditions. There are two main sources of interference; one factor is the interference of the external environment. During the imaging process of the hyperspectral imager, atmosphere absorption of radiation of different wavelengths, sharp changes in light, and photon effects lead to image quality degradation [7], [8]. Another influencing factor comes from the imaging instrument, such as sensor temperature, exposure time, electromagnetic interference, and data transmission. This interference can cause various kinds of noise, for instance, Gaussian noise, impulse noise, stripes, and deadline [9], [10], [11]. Consequently, degraded HSIs have negative effects on subsequent high-level tasks [12]. Alternative solutions to this problem are hardware-based, but such improvements are complicated and costly [7], [8]. As a result, HSI denoising through software remains a critical and practical step before subsequent processing.

However, denoising HSIs faces significant challenges due to the interference of mixed noise. A variety of ways have been studied to address this issue. Specifically, model-driven denoising has been regarded as an ill-conditioned problem [12], [13], [14], [15], [16], [17], [18], [19], [20], [21]. After introducing prior constraints, such as sparse representation (SR) [16], [18], total variation (TV) [12], [16], [22], nonlocal similarity [21], [23], [24], and low-rank (LR) matrix [15], [19], [20], the denoising problem is transformed into a well-posed one, which allows the clarity extraction of hierarchical structure and feature details from the images. However, the complex prior, artificially constructed, makes the iterative optimization difficult and time-consuming. Over the years, with the flourishing of artificial intelligence, data-driven methods have shown significant advantages in denoising [8], [25]. Especially, convolutional neural networks (CNNs) have become dominant players in this field. Methods, such as 2-D [8], [26], [27], [28] and 3-D [25], [29], [30] convolution filters-based, given noisy HSIs with noisy labels and corresponding clean HSIs, learn and train the nonlinear mapping between them. Compared with the model-based approaches, the CNN-based methods, rather than focusing on the degradation mechanism, can achieve end-to-end denoising with promising results given enough samples. However, the convolution filters of these approaches have fixed weights, which learn priors from given training datasets but ignore the internal information of HSIs. Additionally, the networks are poorly interpretable [31], [32], [33] and complex due to the introduction of redundant uncertain parameters.

In recent years, a transformer [34], known for its excellent performance in natural language processing (NLP), has been employed in image restoration [35], [36], [37]. The

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self-attention mechanism can adaptively extract the spatial structure similarity of HSIs [10], [11], [38], [39], [40], which are sparse in spatial and highly correlated in the spectral domain. Moreover, in order to avoid negative denoising effects due to neglecting spectral correlation, the dual-attention mechanism based on spatial–spectral information has been widely adopted [41], [42]. However, most current denoising efforts ignore the modeling of the physical properties of HSIs, which leads to weak generalization and suboptimal efficiency. Furthermore, the issue of increasing computational complexity of transformer-based methods with the increase of spectral abundance still needs to be addressed.

Therefore, we propose a lightweight S^{2}OTAN for mixed noise removal in complex HSIs. First, a shallow feature extractor is designed to mine spatial and spectral features coarsely, which benefits from the joint utilization of high correlation between adjacent pixels and bands. Second, we construct a spatial–spectral triple attention block (SSTAB) that involves three attention layers to explore local information and long-range dependencies in multidimension to enhance local spectral correlation. Among these layers, we introduce a two-layer cascaded attention by constructing attention maps for pixels within and across windows to facilitate the interaction of local and global information within the spatial domain. Additionally, for spectral feature extraction, unlike the original transformer model that focuses globally on the spectral domain, we utilize spectral partitioning operations to enhance the adjacent spectral constraints of HSIs and capture relevant contextual information. Furthermore, residual connections are employed to ensure network stability and reserve shallow features. Finally, a locally enhanced feed-forward (LeFF) network is established for multiscale spatial–spectral feature aggregation to achieve robust mixed noise removal. In conclusion, the main contributions of this article are as follows.

1) Aim at mixed noise suppression and image details restoration, we propose the lightweight S^{2}OTAN, which builds a local–global spatial–spectral relationship in a parallel manner to fully leverage the physical properties of HSIs.
2) To enhance multiscale spatial feature extraction ability while minimizing computational costs, we design hybrid spatial attention by constructing attention maps for pixels within and across windows, which facilitates the interaction of local and global information within the spatial domain.
3) To improve spectral fidelity and lightweight computation, we utilize spectral partitioning operations to enhance the adjacent spectral constraints of HSIs and capture contextual information related to correlations.
4) Extensive experiments conducted on synthetic and real-noisy scenarios demonstrate that the proposed method is robust in handling complex scenarios. It removes all types of noise and preserves image textures without causing spectral distortion.

II. Related Work

A. Model-Based HSI Denoising

HSIs can be perceived as stacks of grayscale images. Consequently, early denoising tasks relied on adapting natural image denoising methods using band-by-band strategies, such as SR over learned dictionaries [43], sparse 3-D transform-domain filter [44], nonlocal means (NLMs) filter [45], and weighted nuclear norm minimization (WNNM) [46]. However, band-by-band denoising strategies often overlook spectral dependence, which limits their ability to fully utilize the intrinsic characteristics of HSIs. To address this, model-driven methods based on handcrafted priors have been widely studied to exploit the spectral properties effectively, including TV [9], [12], [47], [48], nonlocal prior [23], SR [7], [18], [49], [50], [51], and LR [15], [52], [53], [54], [55], [56]. The TV model is an anisotropic model for smoothing images, capable of preserving edge details while effectively removing noise. Furthermore, to leverage spectral physical properties, spatial–spectral TV (SSTV) [9], [12], [47] was introduced to smooth noise in the spectral domain. However, TV models often produce staircase effects and excessive smoothing, losing original texture features in images. Therefore, the development of SR was implemented to optimize recovery results. For instance, Peng et al. [48] combined TV regularization and LR prior to calculating the gradient map subspace sparsity for all bands. Zhang et al. [51] presented group-based SR, and Lu et al. [18] proposed sparse represent spectral–spatial features adaptively. Besides, LR is also a commonly used regularization term in inverse problems. Zhang et al. [15] designed the LR matrix recovery (LRMR) framework to represent HSIs as the 2-D matrix and used the go decomposition algorithm to solve it. Typically, HSIs are viewed as piecewise smooth in the spatial domain and LR subspace in the spectral domain, resulting in good synergy in regularization terms. Following this, He et al. [52] incorporated the LR matrix and TV model to propose TV-regularized LR matrix factorization (LRTV). However, matrix techniques lead to the lack of spatial detail in HSIs, while tensor modeling can provide more abundant details without compromising the spatial structure of HSIs. Inspired by this, Wang et al. [57] proposed the TV-regularized LR tensor-decomposition method (LRTDTV). They employed the Tucker decomposition to represent the global correlation between spectral bands while characterizing the segmented smoothing structure in spatial and spectral domains by anisotropic SSTV regularization. Furthermore, Chang et al. [58] introduced hyper-Laplacian regularized unidirectional LR tensor recovery (LLRT), which models the global spectral structure by hyper-Laplacian prior. He et al. [59] designed a nonlocal meeting (NGmeet) global model according to the HSIs and the spectral subspaces lying in the global LR subspace. Despite the significant development and remarkable performance of model-based methods, the selection of hyperparameters often depends on empirical rules, making them challenging to optimize. Moreover, most model-based methods are suboptimal in efficiency due to the numerous number of iterations required for optimization algorithms.

B. Learning-Based HSI Denoising

Recently, learning-based methods, especially CNNs have succeeded in denoising. Unlike model-based methods, learning-based methods do not require the design of complex handcrafted priors. Instead, they pay attention to end-to-end data by employing the network to nonlinear map from

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noisy and clean HSIs, improving time efficiency significantly. Inspired by natural image processing techniques [8], [26], [27], [28], initial applications of CNN for denoising primarily used 2-D convolution filters to dig spatial information and eliminate noise band by band. The above-mentioned strategies neglect the spectral characteristics in HSIs, leading to the destruction of the original 3-D structure and a reduction in recovery performance. Recognizing the powerful feature extraction capabilities of 3-D convolution in the spatial–spectral domains, it has gradually become a mainstream approach in HSIs denoising. Liu and Lee [29] designed a 3-D atrous denoising CNN, which expands the receptive field without increasing additional parameter burden. Wei et al. [30] designed an alternative directional 3-D quasi-recurrent neural network (QRNN3D) to capture the global correlation of spectra and eliminate causal dependencies. In addition, to overcome the limitations of the high computational complexity of 3-D convolution, Dong et al. [25] decomposed 3-D convolution filters into 1-D and 2-D filters, which reduces parameter number and computational complexity. However, CNN models act as black boxes, which can lead to trouble, such as weak generalization ability, poor interpretability, and unstable performance. To address these, researchers choose to combine CNN with modeling to overall performance. Lin et al. [31] formulated the denoising problem as a constraint matrix decomposition problem of CNN and optimized it by the proximal alternating linearized minimization strategy. Miao et al. [60] used deep priors instead of handcrafted priors to achieve unsupervised denoising. Another way of combining is to embed prior information into end-to-end training of CNN, such as combining spatial LR prior [61], [62], sparsity prior [63], and long-range correlation [30], [64]. Although most problems have been solved, it remains challenging to maintain high performance and flexibility in real-noisy scenes due to the characteristics of CNNs, namely, limited receptive fields and static weights.

C. Vision Transformer

The transformer [34] completely abandons recursion and CNNs, which adopt multihead self-attention (MSA) mechanism and feed-forward neural network (FFN) to learn long-term dependencies and contextual information efficiently. Driven by the remarkable success of transformers in NLP, efforts have been made to integrate transformers into computer vision tasks with the aim of surmounting the limitations of CNNs. Specifically, Ramachandran et al. [38] proposed a stand-alone self-attention network to directly replace 3×3 spatial convolutions with local spatial self-attention layers, achieving better results in visual tasks and verifying the superiority of the transformer. However, Dosovitskiy et al. [35] pointed out that during the coarse stitching process of vision transformer (ViT) images, more attention is paid to global information, resulting in the loss of edge details. Therefore, relevant works on local transformers have emerged. A window sliding strategy was applied in the Swin transformer [36] to extract local details and perform window interaction. A spatially separable self-attention mechanism was used in twins [39] to simulate local–global representations. Yuan et al. [40] proposed outlook attention to extract more refined details. However, the input–output resolution of ViT remains unified, disregarding fine-grained, which leads to computationally expensive processes in high-density image processing tasks. Therefore, pyramid ViT (PVT) [37] was proposed to incorporate nonoverlapping patches to reduce feature number and improve computational efficiency. Similarly, pooling-based ViT [65] and convolutional ViT [66] were proposed to use pooling and convolution to downsample tokens and change resolution.

In recent years, the transformer has quickly developed in image restoration on account of its outstanding performance. For instance, Wang et al. [67] proposed a general U-shaped transformer model, which adopts an encoding-decoding mode to extract multiscale spatial textures. Inspired by the Swin transformer, Liang et al. [68] proposed the SwinIR model to leverage residual modules to explore local spatial self-similarity. However, the computation complexity of the transformer surges quadratically with the resolution, making it intractable in most high-dimension image restoration tasks. Therefore, Zamir et al. [69] proposed restormer to aggregate local and global pixel interactions and filter out insignificant information. Although the restoration task of HSIs can follow the way of RGB images, due to the 3-D characteristics of HSIs, it is essential to pay additional attention to the local correlation of spectra. To this end, Wang et al. [10] proposed a local–global feature-aware transformer-based residual network, which embeds spectral information in shallow feature extraction. Considering the richness of spectral information, Hider [11] constructs a global spectral attention module that can excavate different spectral patterns. Wang et al. [70] employed a spatial–spectral feature extraction module and a residual module based on the Swin transformer to alternately extract spatial and spectral features, supplemented by TV loss to maintain the smoothness of the HSI. Li et al. [41] designed a dual transformer that uses spatial and spectral self-attention in serial to fuse spatial–spectral information. Although the transformer has succeeded in multiple fields, its development in HSI denoising still faces considerable obstacles. The abundance of spectral bands increases the computational overhead of the network, and the incorporation of HSI characteristics into the network architecture remains a relatively unexplored area [71]. To address these problems, this study aims to leverage a 2-D transformer to model the 3-D characteristics and prior knowledge of HSIs, achieving impressive denoising performance while maintaining low computational complexity.

III. Method

A. Formulation

An ideal HSI $\mathbf{X} \in \mathbb{R}^{H\times W\times B}$ degrade into a noisy HSI $\mathbf{Y} \in \mathbb{R}^{H\times W\times B}$ because of the influence of mixed noise $\mathbf{N} \in \mathbb{R}^{H\times W\times B}$, the degradation processing can be represented as

$$\mathbf{Y} = \mathbf{X} + \mathbf{N} \tag{1}$$

where H and W denote the height and width of the HSI, respectively. B represents the spectral dimension of the HSI. HSIs denoising aims to restore the clean HSI $\mathbf{X}$ from the noisy

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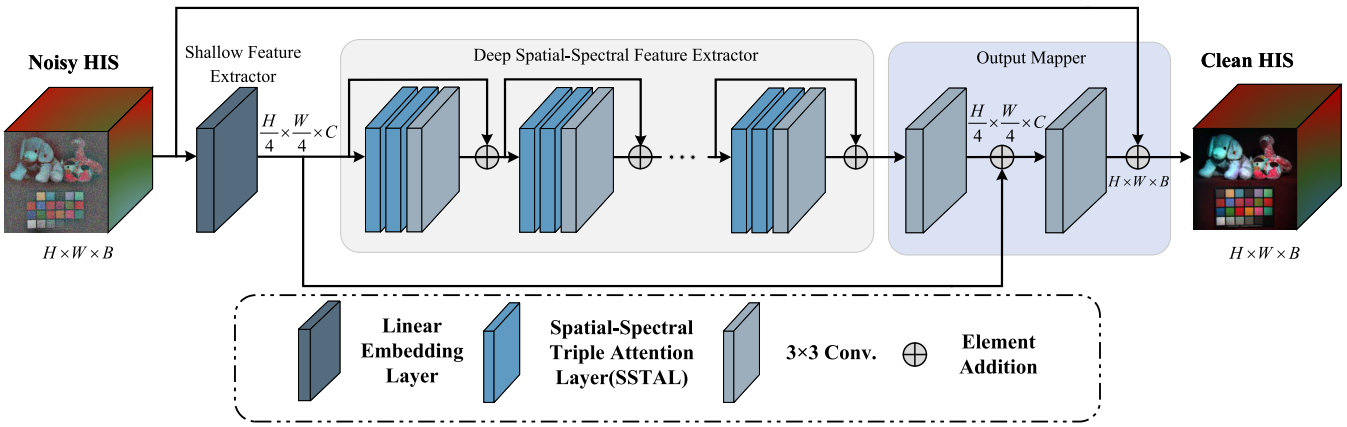


Fig. 1. Overall architecture of the proposed S^{2}OTAN for HSI denoising. The network contains a shallow feature extractor, a deep spatial–spectral feature extractor, and an output mapper.

HSI $\mathbf{Y}$. However, obtaining $\mathbf{X}$ directly through the degradation model is an ill-conditioned inverse problem, which has infinitely many solutions. To solve it, we propose the S^{2}OTAN network $f(\cdot)$ to simulate the noise $\hat{N} = -f(\mathbf{Y})$ through inputting the noisy HSI $\mathbf{Y}$, which can increase convergence and avoid overfitting during network training.

B. Overall Network Architecture

As illustrated in Fig. 1, the proposed network framework contains three modules: shallow feature extractor, deep spatial–spectral feature extractor, and output mapper. First, the degraded HSI is fed into a shallow feature extractor, which divides the HSI into nonoverlapping patches in the spatial domain and transforms it into sequence embeddings. Then, a linear embedding layer is employed to project the original features to a designed dimension. Subsequently, to dig profound feature representations, the original features processed by the shallow feature extractor are further fed into the deep spatial–spectral feature extractor, which consists of SSTABs and convolutional layers. Inspired by the Swin transformer [36], in each SSTAB, we adopt a partitioning strategy in space to dig local features of HSIs and obtain long-range similarity by shift operation. Simultaneously, we divide the adjacent bands of the spectrum and leverage the local correlation to mine spectral information. Subsequently, the fusion is conducted between spatial and spectral features by LeFF [67] network. Additionally, we employ residual connections and layer normalization (LN) [72] to avoid the network from falling into a hierarchy crash. Finally, the shallow and deep features are fused into the output mapper through skip connections to enhance the restoration of HSIs, where the output mapper consists of two 3×3 convolutional layers.

C. Shallow Feature Extractor

HSIs have high spatial similarity and spectral correlation. To leverage this rich inherent information to complete the image restoration task, we utilize a linear embedding layer to accomplish shallow spatial–spectral feature extraction. First, we divide the noisy HSI $\mathrm{Y} \in \mathbb{R}^{H \times W \times B}$ into patches with the size of 4×4 as tokens, thereby the feature dimension of each token turn to $4 \times 4 \times B$, as illustrated in Fig. 2. Then, we project the feature dimension to a specific size

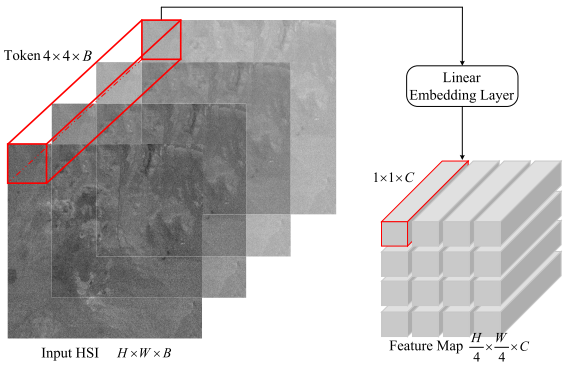


Fig. 2. Details of feature map of shallow extraction using linear embedding layer.

(denoted by C). The advantage of this embedding strategy is that it considers spatial and spectral characteristics and reduces memory cost and calculation overhead. The acquired feature maps with the size of $(H/4) \times (W/4) \times C$ will be utilized for deeper feature extraction.

D. Spatial–Spectral Triple Attention Block

In the denoising of HSIs in the natural world, although we can fully use the long-range spatial dependencies by global spatial self-attention, it may lead to excessively exponential complexity. Additionally, local window self-attention only provides information within the nonoverlapping window, which limits its effectiveness in capturing broader contextual information. To build a local–global spatial–spectral relationship to fully leverage the physical property of HSIs, we introduce the SSTAB, which is composed of two parallel spatial–spectral branches and a spatial–spectral features aggregation network, as shown in Fig. 3. Specifically, the spatial branch contains two cascaded spatial attention layers, i.e., the splitted-window MSA (SW-MSA) layer and the cross-window MSA (CW-MSA) layer. Another spectral branch comprises one inner spectral MSA (IS-MSA) layer.

In terms of the spatial branch, although the attention extraction operation in a small window can reduce the computational cost, the modeling ability of SW-MSA is limited. Therefore, we use the cross-window operation to exchange information between adjacent windows to improve feature extraction performance. This hybrid strategy focuses on both local and

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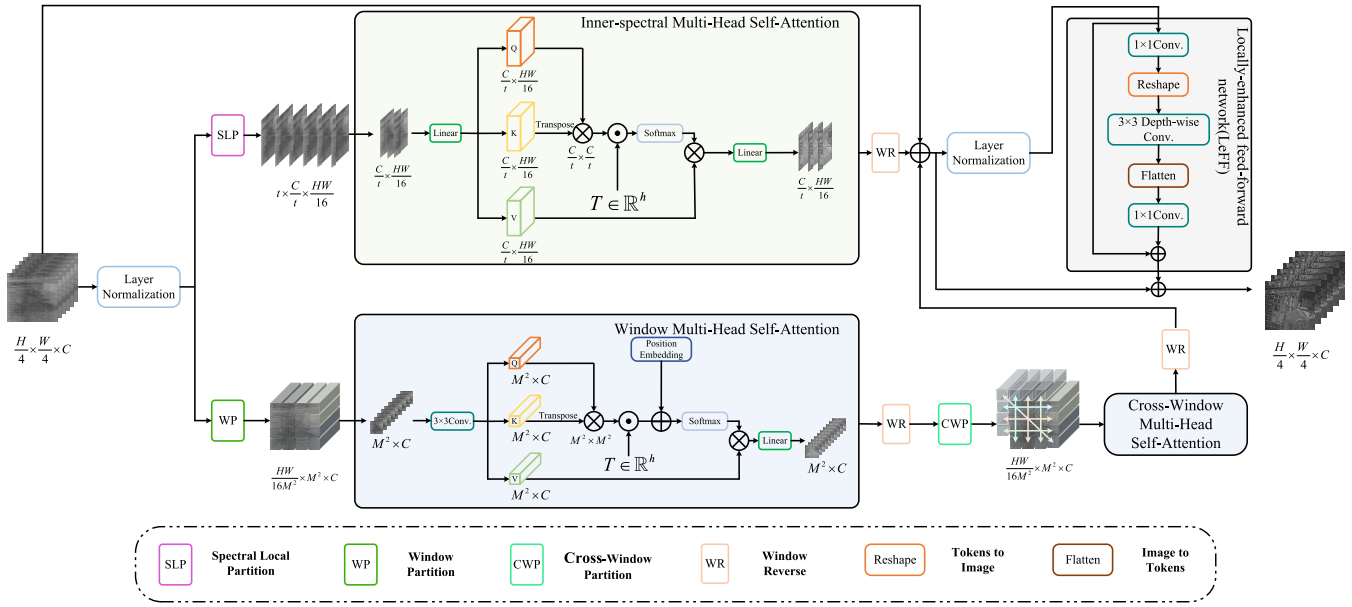


Fig. 3. Overall architecture of the SSTAB. It contains two parallel branches and an LeFF network. The spatial branch consists of two cascaded spatial attention layers, SW-MSA and CW-MSA. The spectral branch comprises an inner spectral attention layer, IS-MSA.

global spatial similarity, which enhances multiscale spatial feature extraction ability while minimizing computational costs.

In terms of the spectral branch, unlike the original transformer model that focuses globally on the spectral domain, we utilize spectral partitioning operations to enhance the adjacent spectral constraints of HSIs and capture contextual information related to correlations. This strategy reduces the computation load and improves feature extraction ability simultaneously.

After extracting local features from the spatial and spectral domains, we employ LN to avoid a hierarchy crash of the network. Subsequently, we transmit the features to LeFF for spatial–spectral feature aggregation.

1) Splitted-Window Multihead Self-Attention: As presented in Fig. 3, after the feature map is normalized, we perform a window partition operation, and then the map will be divided into windows of size $M \times M$ in the spatial dimension. We can obtain $(H \times W/16M^2)$ windows of size $M \times M \times C$. After getting the input ith feature $Z_i \in \mathbb{R}^{M^2 \times C}$, we utilize the linear transformation matrix to project it into query, key, and value matrices, which are represented as Q_{ij}, K_{ij}, and V_{ij}, respectively

$$\begin{aligned} Q_{ij} &= Z_i W_j^Q \\ K_{ij} &= Z_i W_j^K \\ V_{ij} &= Z_i W_j^V \end{aligned} \tag{2}$$

where $Q_{ij}, K_{ij}, V_{ij} \in \mathbb{R}^{M^2 \times d}$. $W_j^Q, W_j^K, W_j^V \in \mathbb{R}^{C \times d}$ are jth shared linear projection matrices. Then, we use a relative position bias $B \in \mathbb{R}^{M^2 \times M^2}$ and a vector $T \in \mathbb{R}^h$ to calculate the self-attention matrix

$$\text{Attention}\left(Q_{ij}, K_{ij}, V_{ij}\right) = \text{Softmax}\left(Q_{ij} {K_{ij}}^T \odot T + B\right) V_{ij} \tag{3}$$

Since the relative position range of each axis is $[-M+1, M-1]$, the values of B are acquired from the bias matrix $\hat{B} \in \mathbb{R}^{(2M-1)\times(2M+1)}$. The attention maps are adjusted by the learned weight vector $T \in \mathbb{R}^h$, which can adaptively adjust the size of the input matrix. Consequently, the input of the Softmax function will be stabilized within a range, and the stability of network optimization will be significantly improved. Then, we concatenate all features of MSA and multiply them with a weight matrix to acquire local spatial attention features

$$\begin{aligned} \text{Attention}(Q_i, K_i, V_i) &= \text{Concat}(\text{head}_1, \text{head}_2, \ldots, \text{head}_h) W^o \\ \text{head}_j &= \text{Attention}\left(Q_{ij}, K_{ij}, V_{ij}\right) \end{aligned} \tag{4}$$

where $W^o \in \mathbb{R}^{hd \times d}$ represents the weight matrix, h represents the number of multihead. The dimension of the query/key d is equal to (C/h).

2) Cross-Window Multihead Self-Attention: Considering the limitations of the local window self-attention, which can only work within the window and lacks interaction between windows, we introduce a cross-window information exchange layer, i.e., CW-MSA, as illustrated in Fig. 3. $\lfloor |(M/2)|, |(M/2)| \rfloor$ pixels are shifted when the input features are partitioned. Thus, the feature information of different windows is integrated. After the shifted transformation, the MSA operation is similar to the SW-MSA layer.

3) Inner Spectral Multihead Self-Attention: After obtaining a partial spectral representation in shallow feature extraction, we present an IS-MSA layer to excavate spectral information at a deeper level. Specifically, given input feature map $Z \in \mathbb{R}^{(H/4)\times(W/4)\times C}$, we transpose and divide it to $Z^T \in \mathbb{R}^{t\times(C/t)\times(HW/16)}$. Subsequently, we calculate the matrix query $Q^g \in \mathbb{R}^{(C/t)\times(HW/16)}$, key $K^g \in \mathbb{R}^{(C/t)\times(HW/16)}$ and value $V^g \in \mathbb{R}^{(C/t)\times(HW/16)}$ as 2. We can express the inner spectral self-attention operation as

$$\text{Attention}\left(Q^g, K^g, V^g\right) = \text{Softmax}\left(Q^g {K^g}^T \odot T\right) V^g. \tag{5}$$

Similar to 4, we feed these matrices into the heads separately. Finally, the resulting matrix is restored to size $(H/4) \times (W/4) \times C$. Through this strategy, global spectral contextual

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information is excavated from noisy HSIs, enabling effective and lightweight recovery.

E. Locally Enhanced Feed-Forward Network

Linear transformation and nonlinear activation are provided by the FFN to enhance the representation power of features in the standard transformer [66], [73]. However, its ability to utilize contextual dependencies is limited, which plays a vital role in image recovery tasks. Based on recent work [67], we replace FFN with the LeFF to achieve correlations between adjacent pixels. The structure of LeFF is shown in Fig. 3, where the input token is first increased to a specific dimension using a 1×1 convolutional layer, then reshaped into 2-D features, and finally local information is captured by a 3×3 depthwise convolution. To this end, the tokens are flattened and contracted back to the original channels. GELU serves as the activation function for each convolutional layer.

IV. Experiments

A. Implementation Details

To verify the superiority of the proposed method for HSIs denoising tasks, it is compared with thirteen competing methods on synthetic and real-noisy datasets, including matrix-based LR methods (LRMR [15] and NMoG [74]), tensor-based LR methods (NGmeet [59] and LLRT [58]), a TV-based method (enhanced 3-D total variation (E3DTV) [48]), TV regularized matrix-based LR methods (LRTV [52] and LRTDTV [57]), CNN-based methods (QRNN3D [30], model-aided nonlocal neural network (MAC-Net) [32], and nonlocal spatial–spectral neural network (NSSNN) [33]), and transformer-based methods (3-D quasi-recurrent and transformer based network (TRQ3D) [75], SST [41], and SERT [71]).

1) Dataset: The Columbia Imaging and Vision Laboratory (CAVE) dataset [76] contains 32 HSIs, of which 26 are selected for training. Specifically, images of size $512 \times 512 \times 31$ are cropped into patches of size $64\times64\times31$ with a stride of (16, 16). Data augmentation techniques, such as linearization, rotation, and flipping, are employed to enrich the dataset, resulting in approximately 50 000 3-D image patches. The remaining data are used for validation and testing without any cropping or resizing. In addition, to further validate the robustness of the algorithm in different scenarios, Chikusei [77] and Pavia University datasets are also applied in simulation experiments. To evaluate the denoising performance on real-world noise, the Urban, Cuprite, and Indian Pines remote sensing datasets are selected. What is noteworthy is that due to limited computational resources and significant differences in scene and parameters among different remote sensing datasets, fine-tuning is infeasible. However, it is feasible to denoise the remote sensing images by using the model trained on the CAVE dataset. To address differences in spatial resolution and spectral bands, the images are split into multiple 31-channel subimages, and the denoising results are aggregated to generate denoised HSIs. Descriptions of the relevant datasets are provided in the following.

1) *CAVE*: It was taken by a cooled charge-coupled device (CCD) camera (Apogee Alta U260) and covers 31 bands from 400 to 700 nm. The images have a size of $512 \times 512 \times 31$, with 26 of them used as training samples and the rest as test samples.
2) *Chikusei*: It consists of 128 bands, with the wavelength range from 363 to 1018 nm, acquired by Canon EOS 5D Mark II. The spatial image is composed of 2517×2335 pixels. We crop it into 512×512 and add mixed noise for synthetic denoising experiments.
3) *Pavia University*: It was collected in Pavia, Italy, using the imaging system ROSIS-03. It comprises 610×340 pixels and 115 bands, which were cropped into 256×256. Mixed noise was added to these patches for synthetic denoising experiments.
4) *Urban*: HYDICE sensor captured it with 307×307 pixels and 210 bands, wavelength range from 400 to 2500 nm. We crop it into 256×256 for real-denoising experiments.
5) *Cuprite*: It was acquired by Aviris and consists of a size of 512×614 and 224 bands, ranging from 370 to 2480 nm. We crop it to 512×512 for real-denoising experiments.
6) *Indian Pines*: It was obtained in Indiana, USA, and we retain all the original 220 bands and crop it into 128×128 for real-denoising experiments.

2) Quantitative Metrics: Four standard metrics are adopted to accurately assess the denoising performance, including the peak signal-to-noise ratio (PSNR), the structural similarity (SSIM) [78], the spectral angle mapper (SAM) [79], and relative dimensionless global error in synthesis (ERGAS) [80]. Among them, the larger the PSNR and SSIM values, the higher the similarity between the denoising image and the original image. The smaller the SAM and ERGAS values, the higher the fidelity of the denoising image.

3) Noise Setting: In the synthetic experiment, to better simulate the noise in the real world, we artificially add a variety of noises to CAVE in six cases, including Gaussian, impulse, deadline, and stripe noise, as follows.

1) *Case 1 [Independent and Identically Distributed (i.i.d.) Gaussian Noise]*: Add white Gaussian noise with zero-mean to each band evenly to simulate the synthetic data, and set the variance to four subcases (Case 1.1–1.4 and $\sigma = \{10, 30, 50, 70\}$).
2) *Case 2 (Non-i.i.d. Gaussian Noise)*: Various bands are corrupted by Gaussian noise with various noise variances randomly set in the range of 10–70.
3) *Case 3 (Non-i.i.d. Gaussian Noise + Deadline Noise)*: Based on Case 2, deadline noise of length 5%–15% of columns is randomly distributed on one-third of the band.
4) *Case 4 (Non-i.i.d. Gaussian Noise + Stripe Noise)*: Similar to Case 3, replace deadline noise with stripe noise.
5) *Case 5 (Non-i.i.d. Gaussian Noise + Impulse Noise)*: Based on Case 2, randomly selected one-third of the bands are simulated with impulse noise with 10%–70% intensities.
6) *Case 6 (Mixed Noise)*: Combine non-i.i.d. Gaussian, deadline, stripe, and impulse noise in the above-mentioned cases.

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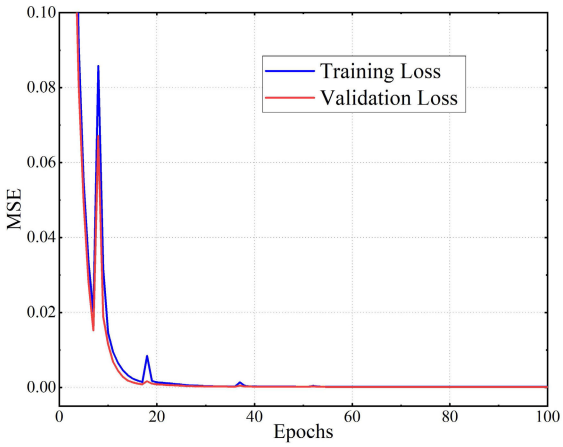


Fig. 4. Training and validation convergence curves of S^{2}OTAN.

B. Training Details

To optimize the proposed model, we employ the Adam optimizer [81] with a learning rate of 1×10^{-4}, which is changed to 1×10^{-5} after 60 epochs. A total of 100 epochs are trained with a batch size of 4. Mean squared error (MSE) is chosen as the loss function. The training dataset is prepared as described in Section IV-A1. To accommodate various noise types, different noise settings are used for training. In scenarios involving pure Gaussian noise (Case 1.1–1.4 and Case 2), noise settings similar to Case 2 are applied to process the training set. For complex noise combining multiple noise types and real data, noise settings similar to Case 6 are used. In the shallow feature extraction module, the embedding dimension is set to 90. Sensitivity analysis on the embedding dimension will be presented in the ablation experiments. In the deep feature extraction module, a residual module contains six SSTABs, with six attention heads. The window size for SW-MSA and CW-MSA is set to 8, and the parameter t for IS-MSA is set to 10. Our implementation is carried out on the PyTorch 1.9.0 deep learning framework, using a platform with an Intel Xeon Processor E5-2650 v4, 64-GB RAM, and NVIDIA GeForce RTX 1080Ti (10 GB) GPU. The other model-based methods are implemented on a platform with an Intel Core i5-8400 CPU and 16-GB RAM. The training and validation convergence curves of the synthetic experiments are shown in Fig. 4 to demonstrate the stability and convergence of the training process.

C. Synthetic Experiments

To verify the algorithm can retain the spectra while denoising, the restoration results of pseudocolor form with the band (31, 11, 6) under mixed noisy setting Case 1.3 are presented in Fig. 5. As we can see, LRMR, LRTDTV, and NMoG do not perform well, as there is still a significant amount of Gaussian white noise in the images. The E3DTV method, which employs TV for denoising, successfully removes the Gaussian noise, but the imposed constraints result in over-smoothing and blurring effects. Additionally, due to the lack of spectral restoration ability, the entire image appears darker. This phenomenon also occurs in the denoising images of LRTV. Due to the LR tensor methods aligning with the characteristics of HSIs, LLRT, and NGmeet outperform other model-based methods. However, LLRT emanates artifacts in the image center and the toy's head, while NGmeet removes noise at the expense of losing edge textures. By contrast, several learning-based methods effectively preserve details with the help of data prior. However, based on the performance of pseudocolor images, it can be observed that the spectral recovery of SST lacks precision. QRNN3D tends to excessively process texture details, leading to slight blurring. The results of MAC-Net still exhibit some residual noise. TRQ3D, NSSNN, SERT, and the proposed method balance noise suppression and detail restoration, achieving the best visual performance. More intuitively, Fig. 6 illustrates the error between the results of competing approaches in the first band and the original images. Compared with most methods that produce deviations between restored and original images, our method achieves the slightest error, demonstrating the closest resemblance between the restored and original images.

Figs. 7 and 8 display the denoising results of competitive algorithms under mixed noisy settings in Case 6 and error maps, separately. From Fig. 7(a) and (b), it is seen that the mixed noise mainly contains impulse noise, deadlines, stripe noise, and Gaussian noise. Model-based denoising methods have limited capabilities for complex noise scenes containing structural noise, such as deadline and stripe noise. LRTDTV maintains deadlines and stripes in the results, while a mass of Gaussian white noise and impulse noise still remains in the denoised images of LRMR and NMoG, resulting in whitened images. Due to the smoothing properties of TV and LR, E3DTV, LRTV, LLRT, and NGmeet all blur results to varying degrees. Additionally, E3DTV and NGmeet fail to retain the spectra, showing a significantly darker image background. Owning to the powerful spatial–spectral extraction capability of 3-D convolution, QRNN3D can effectively preserve details. However, its ability to suppress noise is limited, resulting in suboptimal spectral fidelity. The introduction of prior knowledge enables MAC-Net to produce images with clear textures and accurate spectral recovery, although some residual noise remains. On the other hand, SST, TRQ3D, and SERT suffer from significant loss of background information. In contrast, NSSNN and our method consistently achieve precise spectral restoration. To better present the degree of consistency between the denoised images and the original ones, we plot spectral digital number (DN) values curve at pixel (150, 200) for each band. As shown in Fig. 9, each method deviates from the original curve to varying degrees, but our approach most closely follows the original curve.

In addition to visual evaluation, we also use quantitative metrics to verify the superiority of the proposed method on the CAVE dataset, as displayed in Table I with bold font indicating the best results, while the underlined font is the second-best. Under i.i.d. Gaussian noise interference, both model-based and learning-based methods are effective in denoising, whereas model-based methods perform even better with low-intensity Gaussian noise. However, learning-based methods are more advantageous and faster under high-intensity noise and complex structural noise interference. The proposed method achieves optimal or near-optimal results across

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TABLE I
QUANTITATIVE EVALUATION OF DENOISING RESULTS IN THE CAVE DATASET

Index	Noisy	E3DTV	LLRT	LRMR	LRTDTV	LRTV	NGmeet	NMoG	QRNN3D	TRQ3D	SST	MAC-Net	NSSNN	SERT	S^2OTAN
Case 1.1~1.4: i.i.d. Gaussian Noise															
PSNR	28.13	35.97	42.16	37.23	43.40	31.02	**46.40**	35.43	40.49	37.69	38.47	39.12	39.26	44.61	45.98
SSIM	0.7834	0.9569	0.9931	0.9553	0.9894	0.8883	0.9960	0.9565	0.9955	0.9857	0.9958	0.9692	0.9800	0.9994	**0.9995**
SAM	0.7756	0.3342	0.1539	0.5100	0.1545	0.1393	**0.0773**	0.5021	0.2037	0.1888	0.1791	0.2196	0.1649	0.1213	0.1117
ERGAS	185.63	105.72	23.74	65.38	33.71	133.35	**22.95**	81.62	67.22	80.80	67.93	36.50	58.71	28.77	28.00
PSNR	18.59	31.74	39.98	28.64	37.26	30.08	41.29	28.62	38.61	36.73	36.91	36.09	37.37	41.65	**42.55**
SSIM	0.3560	0.9129	0.9841	0.8627	0.9320	0.8841	0.9848	0.8287	0.9947	0.9802	0.9944	0.9190	0.9678	0.9988	**0.9991**
SAM	1.0463	0.3861	0.1948	0.8425	0.3629	0.1813	**0.1306**	0.7671	0.2409	0.2152	0.2363	0.3180	0.1929	0.1602	0.1328
ERGAS	557.07	144.50	50.87	174.86	65.67	136.77	41.47	177.15	74.48	85.83	79.55	66.78	67.94	**39.84**	40.15
PSNR	14.15	29.40	38.93	25.05	32.96	30.42	38.71	25.42	37.14	35.80	35.70	35.68	36.55	39.38	**40.30**
SSIM	0.2031	0.8849	0.9747	0.8084	0.8350	0.8760	0.9737	0.7175	0.9937	0.9712	0.9928	0.8367	0.9562	0.9980	**0.9984**
SAM	1.1617	0.3967	0.1766	0.9528	0.5589	0.2442	**0.1602**	0.8692	0.2880	0.2630	0.2688	0.4383	0.2185	0.2084	**0.1602**
ERGAS	928.31	178.55	55.26	264.19	107.33	142.75	55.57	255.35	82.60	93.21	90.47	87.35	75.96	51.35	**50.83**
PSNR	11.23	28.05	37.23	22.38	29.84	30.38	34.65	23.36	35.79	34.92	34.68	33.33	35.84	37.45	**38.45**
SSIM	0.1331	0.8636	0.9163	0.7878	0.7047	0.8682	0.9375	0.6287	0.9924	0.9616	0.9912	0.9452	0.9459	0.9969	**0.9975**
SAM	1.2341	0.4098	0.3145	1.0184	0.6943	0.2889	0.2515	0.9311	0.3494	0.3060	0.2901	0.5325	0.2421	0.2690	**0.1627**
ERGAS	1300.08	204.53	115.69	359.23	153.54	143.29	88.11	324.52	91.42	101.19	100.39	109.04	83.62	63.81	**61.63**
Case 2: Non-i.i.d. Gaussian Noise															
PSNR	16.49	32.26	34.93	25.74	33.13	34.37	32.62	29.03	38.57	35.81	35.89	38.65	40.04	41.67	**42.90**
SSIM	0.2136	0.9031	0.9304	0.7672	0.7759	0.9333	0.8806	0.7652	0.9766	0.9463	0.9616	0.9445	0.9643	0.9933	**0.9952**
SAM	1.2977	0.4910	0.2316	1.1901	0.6903	0.2192	0.3569	1.0526	0.3444	0.3585	0.3798	0.3098	0.2161	0.1899	**0.1531**
ERGAS	1455.02	196.59	133.03	597.82	217.03	142.83	199.52	430.69	99.25	170.93	144.35	89.06	81.61	61.67	**53.84**
Case 3: Non-i.i.d. Gaussian Noise + Deadlines															
PSNR	18.41	32.34	35.27	26.47	33.15	34.18	32.04	32.75	39.49	35.99	35.87	40.46	40.43	42.21	**43.17**
SSIM	0.3010	0.9090	0.9245	0.7955	0.7813	0.9248	0.8603	0.8882	0.9820	0.9460	0.9595	0.9637	0.9699	0.9895	**0.9954**
SAM	1.2819	0.4391	0.2307	1.2088	0.7458	0.2930	0.5133	0.6114	0.2833	0.3513	0.3529	0.2543	0.2269	0.1787	**0.1518**
ERGAS	1361.46	200.63	143.02	638.25	233.72	147.16	210.41	176.87	87.56	177.47	153.06	73.11	83.93	65.69	**52.59**
Case 4: Non-i.i.d. Gaussian Noise + Stripes															
PSNR	18.36	32.59	33.27	26.46	33.96	34.21	32.02	30.99	39.01	35.51	35.17	40.01	40.08	41.73	**43.25**
SSIM	0.2983	0.9095	0.9236	0.7931	0.8001	0.9254	0.8507	0.8194	0.9757	0.9553	0.9587	0.9609	0.9715	0.9868	**0.9956**
SAM	1.2814	0.4522	0.2725	1.2167	0.6905	0.2854	0.5466	1.0507	0.3410	0.3154	0.3576	0.2604	0.2074	0.1728	**0.1496**
ERGAS	1360.85	189.24	152.93	611.67	212.61	146.07	208.93	516.92	104.17	156.81	148.00	77.41	84.42	62.83	**51.66**
Case 5: Non-i.i.d. Gaussian Noise + Impulse Noise															
PSNR	18.96	33.17	34.98	27.07	33.99	34.45	32.68	31.25	39.32	35.85	35.95	39.82	40.46	42.17	**43.41**
SSIM	0.3228	0.9138	0.9454	0.8031	0.8017	0.9305	0.8869	0.8205	0.9756	0.9493	0.9617	0.9682	0.9483	0.9884	**0.9955**
SAM	1.2744	0.4765	0.2218	1.1943	0.6866	0.2313	0.3417	1.0200	0.3594	0.3751	0.3632	0.2873	0.2142	0.1779	**0.1482**
ERGAS	1332.01	181.55	134.72	655.75	215.81	143.14	197.32	414.63	103.18	171.49	145.08	82.95	89.29	65.59	**51.02**
Case 6: Mixed Noise															
PSNR	19.43	32.75	35.12	27.45	34.76	34.31	32.04	29.52	39.99	36.03	35.79	40.21	41.15	42.23	**43.38**
SSIM	0.3343	0.9143	0.9344	0.7342	0.8330	0.9243	0.8666	0.7544	0.9808	0.9583	0.9643	0.9638	0.9755	0.9871	**0.9955**
SAM	1.2589	0.4590	0.2159	1.2060	0.6388	0.2833	0.5196	1.1933	0.2827	0.2834	0.3210	0.2707	0.2121	0.1904	**0.1487**
ERGAS	1210.42	188.12	138.21	653.41	204.34	145.42	213.97	689.19	91.97	158.02	143.71	74.59	74.48	63.68	**51.19**

multiple indicators and various environments, consistent with the aforementioned visual evaluation results. Additionally, our results achieve the top performance in PSNR and SSIM among spectral bands, as presented in Fig. 10, which manifests that our method is effective on the overall dataset and in each band.

Furthermore, to evaluate the robustness of our method, the remote sensing datasets Chikusei and Pavia University are selected as supplements. These two datasets are corrupted by mixed noise Case 6 and denoised by competing methods, and denoised images are presented in Figs. 11 and 12, respectively.

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TABLE II
QUANTITATIVE EVALUATION OF DENOISING RESULTS IN CASE 6 OF THE CHIKUSEI DATASET

Index	Noisy	E3DTV	LLRT	LRMR	LRTDTV	LRTV	NGmeet	NMoG	QRNN3D	TRQ3D	SST	MAC-Net	NSSNN	SERT	S^{2}OTAN
PSNR	16.73	30.98	32.28	30.31	31.98	30.73	33.43	34.79	35.21	34.13	33.76	35.24	35.33	34.00	**35.44**
SSIM	0.3141	0.9088	0.9429	0.9057	0.9525	0.8943	0.9587	0.9796	0.9910	0.9733	0.9805	0.9610	0.9621	0.9880	**0.9920**
SAM	1.2821	0.0782	0.0726	0.0927	0.0918	0.0946	0.0831	0.0445	0.0548	0.0730	0.0777	0.0420	**0.0386**	0.0632	0.0435
ERGAS	1345.32	83.48	79.99	84.22	82.50	79.61	55.06	45.71	63.73	98.31	110.29	**32.96**	40.58	76.95	48.23

TABLE III
QUANTITATIVE EVALUATION OF DENOISING RESULTS IN CASE 6 OF THE PAVIA UNIVERSITY DATASET

Index	Noisy	E3DTV	LLRT	LRMR	LRTDTV	LRTV	NGmeet	NMoG	QRNN3D	TRQ3D	SST	MAC-Net	NSSNN	SERT	S^{2}OTAN
PSNR	18.22	32.60	32.83	36.76	36.30	34.06	35.95	37.94	37.44	36.50	37.05	38.24	38.30	37.24	**38.47**
SSIM	0.3219	0.9360	0.9310	0.9453	0.9579	0.9586	0.9536	0.9679	0.9922	0.9860	0.9908	0.9610	0.9808	0.9882	**0.9919**
SAM	1.2453	0.0424	0.0466	0.0592	0.0374	0.0386	**0.0347**	0.0384	0.0452	0.0559	0.0520	0.0432	0.0451	0.0606	0.0419
ERGAS	1310.47	75.46	74.10	45.32	47.65	42.04	48.00	47.23	44.16	55.20	50.48	42.06	43.99	53.88	**41.44**

Local areas of results are zoomed in and placed in the lower left to display textures more accurately.

For the Chikusei dataset, the results of LRMR, LRTDTV, and NMoG still display some Gaussian and impulse noise. Additionally, LRTDTV produces blocky blurring in the paddy field area. With the help of more prior information, E3DTV and LLRT can eliminate most of the noise at the cost of oversmoothing the image, especially E3DTV, which exhibits significant block effects in the house area. The results of LRTV have been blurred to the point where features cannot be distinguished, and a large amount of information is lost. In contrast, both NGmeet and all the tested learning-based methods, including our approach, perform remarkably well, effectively removing noise while maintaining precise details and image structure.

For the Pavia University dataset, the scenes are complex with many trees and buildings as interference, which makes it challenging for model-based methods to balance detail restoration and noise suppression due to the abundance of detail information. As a result, the denoised images of LRMR, LRTDTV, and NMoG all retain a lot of Gaussian noise and stripes. While LLRT, LRTV, and NGmeet, which use LR tensors, all produce different levels of blurring and loss of image information. Although E3DTV has an impressive denoising effect and retains complete edge information, it leads to spectral distortion and whitening of the overall image. Owing to the 3-D convolution extractor, TRQ3D causes oversmoothing at the rooftop, appearing relatively blurry. Among learning-based methods, SST and SERT preserve incomplete edge information and lack spectral fidelity. The details preservation of MAC-Net remains consistently good, although there is still a small amount of residual noise. In comparison, NSSNN, QRNN3D, and our method can achieve both spectral fidelity and structural integrity while removing noise under mixed noise and complex backgrounds.

Quantitative analysis of two remote sensing datasets is presented in Tables II and III. Although there is a slight performance loss, most indicators still outperform the comparative methods. It proves the remarkable robustness of our method without one-to-one trained models, although the significant differences between the imaging environments of remote sensing data and indoor scenes. Furthermore, the complexity and the time consumption of competing methods are shown in Table IV. Model-based methods lack parameter count indicators and run on CPU, while learning-based methods run on GPU, with a test data size of $512 \times 512 \times 31$. It is seen that the proposed transformer-based approach exhibits average levels of parameter count and time, which serves as a direction for future improvement.

D. Real Experiments

To further assess the robustness of the proposed method in complex remote sensing scenarios, we proceed with real experiments on commonly used real-noisy remote sensing images, namely, Urban, Cuprite, and Indian Pines.

1) Urban Dataset: As presented in Fig. 13(a), the image of band 105 is heavily polluted with stripes, deadlines, and random Gaussian noise, making it a challenging denoising task. Most model-based methods struggle to recover clean images, and although LLRT and LRTV remove most of the stripes, they still smooth out important image details. Due to the definition of 3-D TV prior to the clean image, E3DTV produces relatively clean results, but there is also a dynamic range imbalance, leading to local overexposure or underexposure. The dense structural noise also poses a considerable challenge for learning-based methods. Combining the characteristics of both CNN and transformer, TRQ3D introduces dependencies locally and globally to preserve image textures. However, it still exhibits some stripe noise in the dense forest area in the middle of the image. SERT exhibits some excessive smoothing in the complex texture regions, resulting in local blurring. NSSNN shows noticeable sharpening artifacts. QRNN3D fails to completely eliminate structured stripe noise. In contrast, MAC-Net, SST, and the proposed approach effectively eliminate mixed noise and deliver better detail preservation while retaining the original structure.

2) Cuprite Dataset: Taken from a mining area, the Cuprite dataset is a more typical dataset with shaper details and more

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TABLE IV
COMPARISON OF PARAMETERS AND TIME CONSUMPTION OF DIFFERENT METHODS IN CASE 6 OF CAVE DATASET

Index	Model-based methods ($512 \times 512 \times 31$, CPU)							Learning-based methods ($512 \times 512 \times 31$, GPU)						
	E3DTV	LLRT	LRMR	LRTDTV	LRTV	Ngmeet	NMoG	QRNN3D	TRQ3D	SST	MAC-Net	NSSNN	SERT	S^2OTAN
Params (M)	-	-	-	-	-	-	-	0.83	<u>0.68</u>	4.14	**0.43**	1.78	1.91	3.09
Time (s)	45.34	2921.51	64.13	99.94	267.02	244.69	147.86	12.50	15.09	<u>7.41</u>	22.67	14.53	**5.17**	8.35

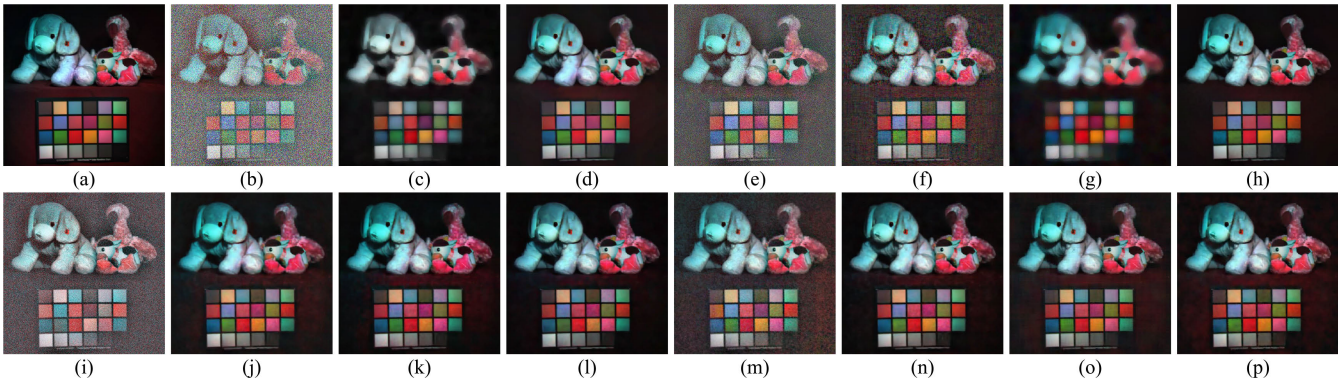


Fig. 5. Synthetic Gaussian noise with noise level $\sigma = 50$ denoising comparison in Case 1.3 of the CAVE dataset, pseudocolor image with bands (31, 11, 6). (a) Original image. (b) Noisy image. (c) E3DTV. (d) LLRT. (e) LRMR. (f) LRTDTV. (g) LRTV. (h) NGmeet. (i) NMoG. (j) QRNN3D. (k) TRQ3D. (l) SST. (m) MAC-Net. (n) NSSNN. (o) SERT. (p) S^2OTAN.

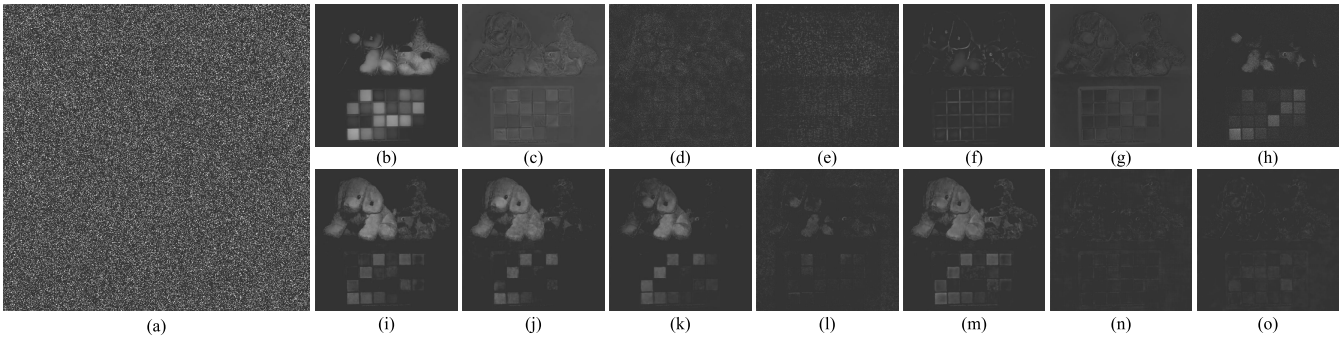


Fig. 6. Band 1 of error map in Case 1.3 of the CAVE dataset. (a) Noisy image. (b) E3DTV. (c) LLRT. (d) LRMR. (e) LRTDTV. (f) LRTV. (g) NGmeet. (h) NMoG. (i) QRNN3D. (j) TRQ3D. (k) SST. (l) MAC-Net. (m) NSSNN. (n) SERT. (o) S^2OTAN.

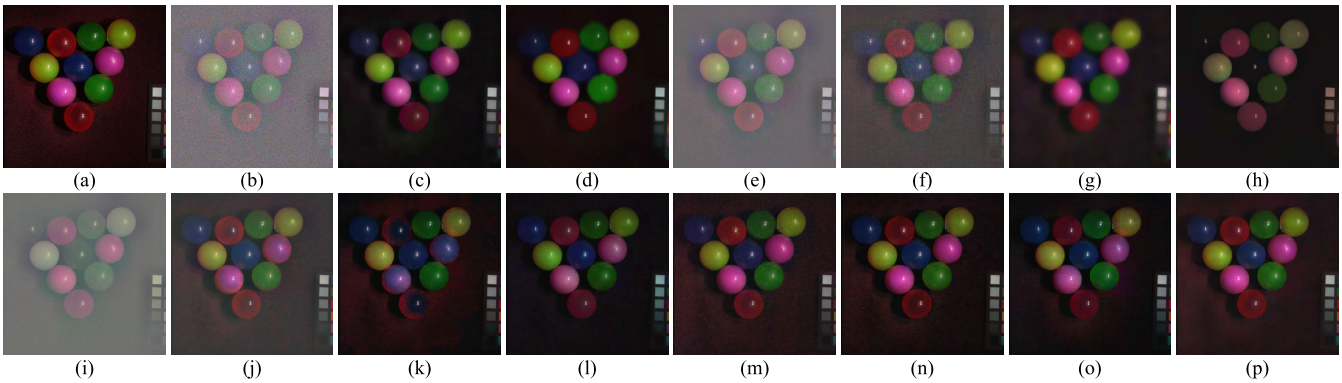


Fig. 7. Synthetic mixed noise denoising comparison in Case 6 of the CAVE dataset, pseudocolor image with bands (31, 11, 6). (a) Original image. (b) Noisy image. (c) E3DTV. (d) LLRT. (e) LRMR. (f) LRTDTV. (g) LRTV. (h) NGmeet. (i) NMoG. (j) QRNN3D. (k) TRQ3D. (l) SST. (m) MAC-Net. (n) NSSNN. (o) SERT. (p) S^2OTAN.

minor gradient differences between edges, making it easy to oversmoothing. Thus, the image of band 223 is selected to evaluate the denoising capability. As shown in Fig. 14, LRMR and NMoG have limited denoising abilities, as there are still Gaussian noise residues in the images. LRTDTV also performs poorly in removing dense structural noise. After further improving the prior constraints, the textures in the image are lost, leading to blurred results of LLRT, LLRTV, and NGmeet. E3DTV performs best among model-based algorithms, which removes most of the noise and only

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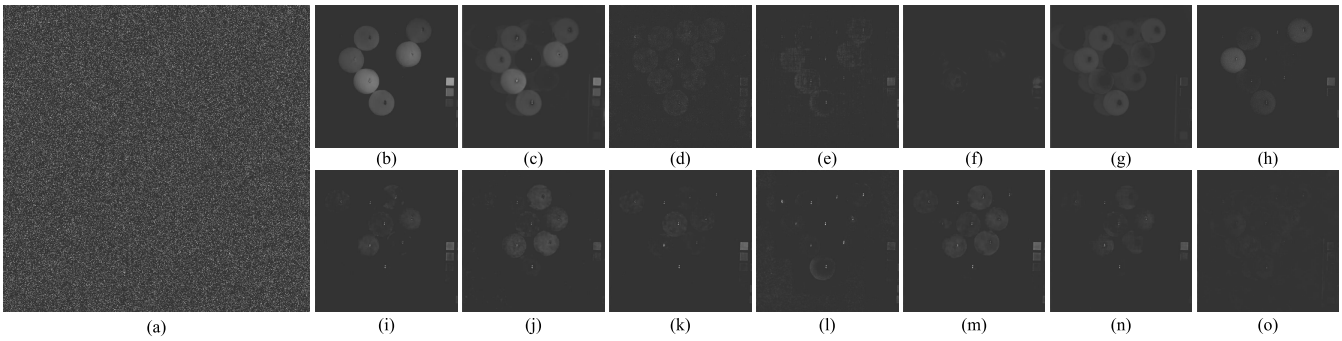


Fig. 8. Band 1 of error map in Case 6 of the CAVE dataset. (a) Noisy image. (b) E3DTV. (c) LLRT. (d) LRMR. (e) LRTDTV. (f) LRTV. (g) NGmeet. (h) NMoG. (i) QRNN3D. (j) TRQ3D. (k) SST. (l) MAC-Net. (m) NSSNN. (n) SERT. (o) S^{2}OTAN.

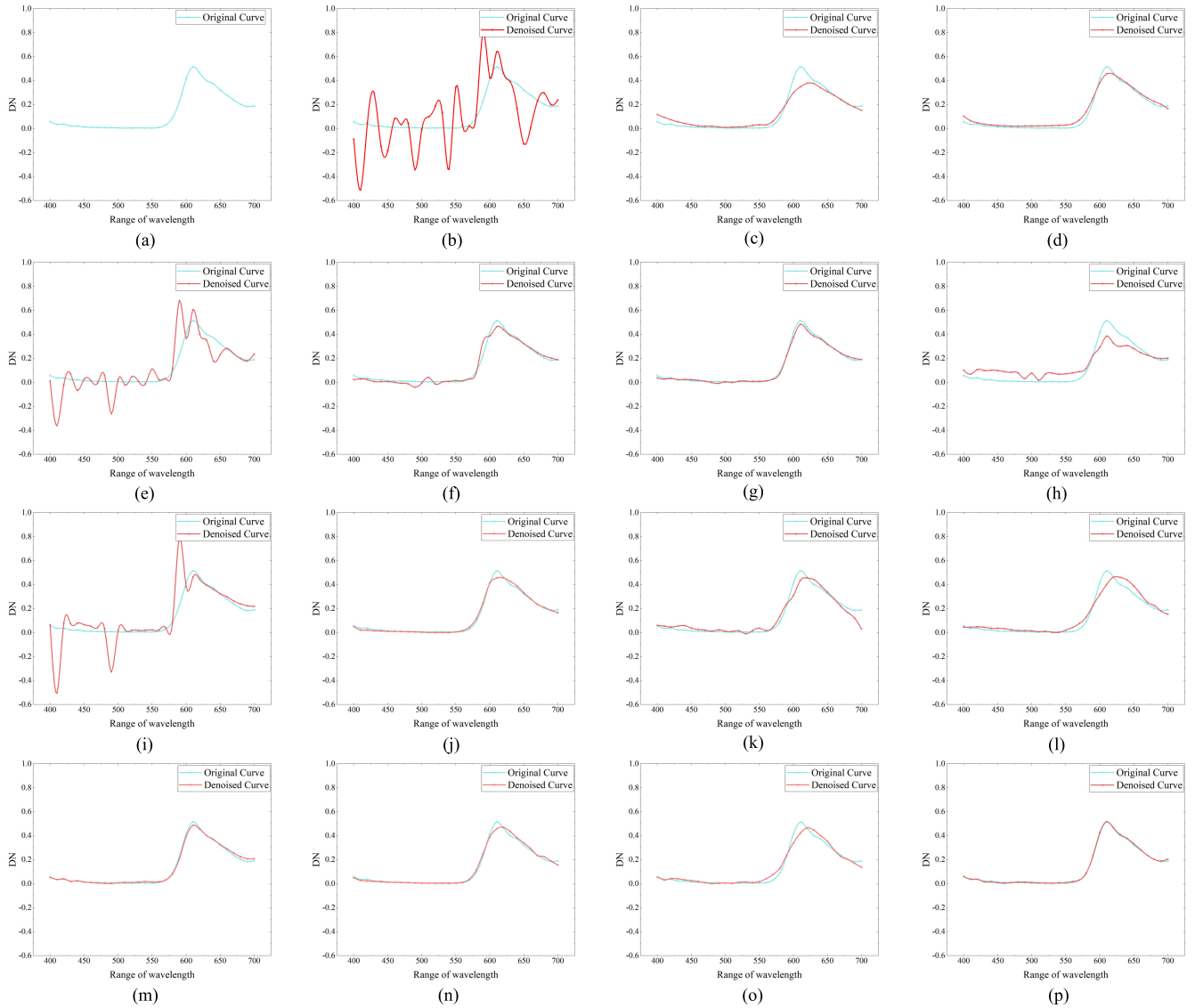


Fig. 9. DN value curves in different bands of pixel (150, 200) after restored from synthetic mixed noise of Case 6. (a) Original image. (b) Noisy image. (c) E3DTV. (d) LLRT. (e) LRMR. (f) LRTDTV. (g) LRTV. (h) NGmeet. (i) NMoG. (j) QRNN3D. (k) TRQ3D. (l) SST. (m) MAC-Net. (n) NSSNN. (o) SERT. (p) S^{2}OTAN.

smoothing features in areas with small image gradients. Both SST and SERT focus on local spatial similarity by adopting a partitioning strategy, which generates the loss of edge details in the final result. MAC-Net benefits from the incorporation of spatial local and nonlocal multiscale prior information, resulting in a highly clear texture in the output. Nevertheless, the noise abatement is not entirely complete. The powerful representation ability of 3-D convolution enables TRQ3D and QRNN3D to improve spectral extraction capabilities while reducing image smoothing. Furthermore, NSSNN and the pro-

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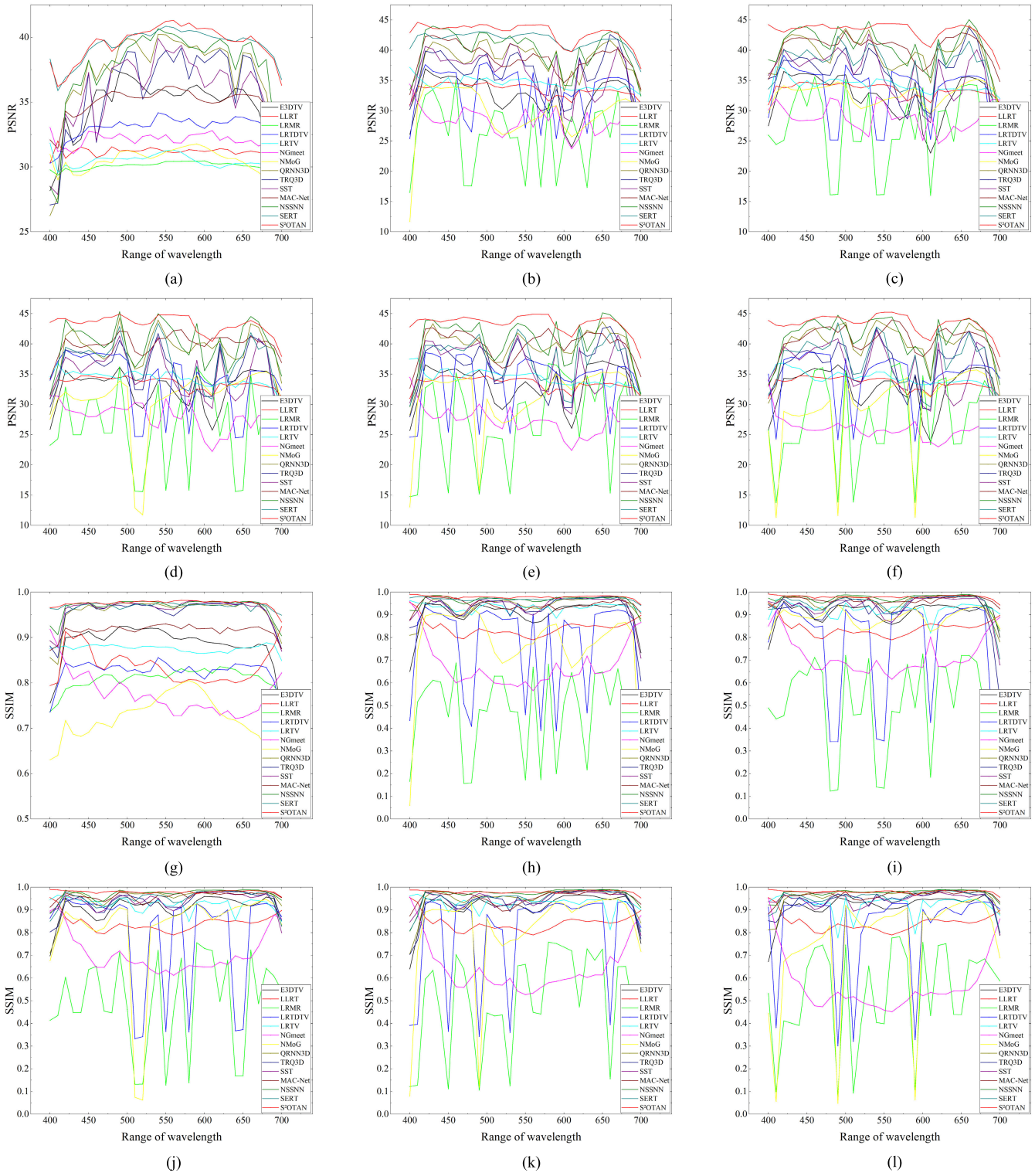


Fig. 10. Comparison of PSNR and SSIM values for different methods across various bands in the CAVE dataset. (a)–(f) and (g)–(l) are PSNR and SSIM, respectively, values of Case 1.3, Case 2, Case 3, Case 4, Case 5, and Case 6.

posed method stand out by combining global spatial structure similarity and local spectral correlation to achieve the best performance.

3) Indian Pines Dataset: Gaussian and impulse noise severely degrades the first few bands of the Indian Pines dataset. To access the denoising efficacy of our method in complex remote sensing scenarios, Fig. 15 displays the grayscale image of band 3 and magnified local areas of results to explicitly contrast details of results of competing algorithms. Although LLRT, LRMR, and NGmeet eliminate most of the Gaussian and impulse noise, the vertical stripes remain unaffected. E3DTV, LRTDTV, and NMoG remove noise under TV prior constraints at the expense of producing artifacts. Nevertheless, further increasing the constraints leads

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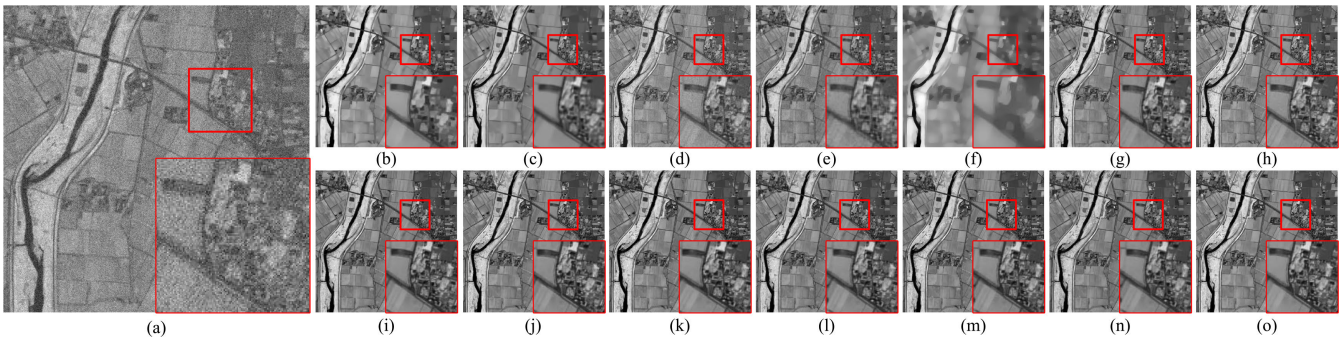


Fig. 11. Denoising results for Chikusei dataset with mixed noise, grayscale display with band 98. (a) Noisy image. (b) E3DTV. (c) LLRT. (d) LRMR. (e) LRTDTV. (f) LRTV. (g) NGmeet. (h) NMoG. (i) QRNN3D. (j) TRQ3D. (k) SST. (l) MAC-Net. (m) NSSNN. (n) SERT. (o) S^2OTAN.

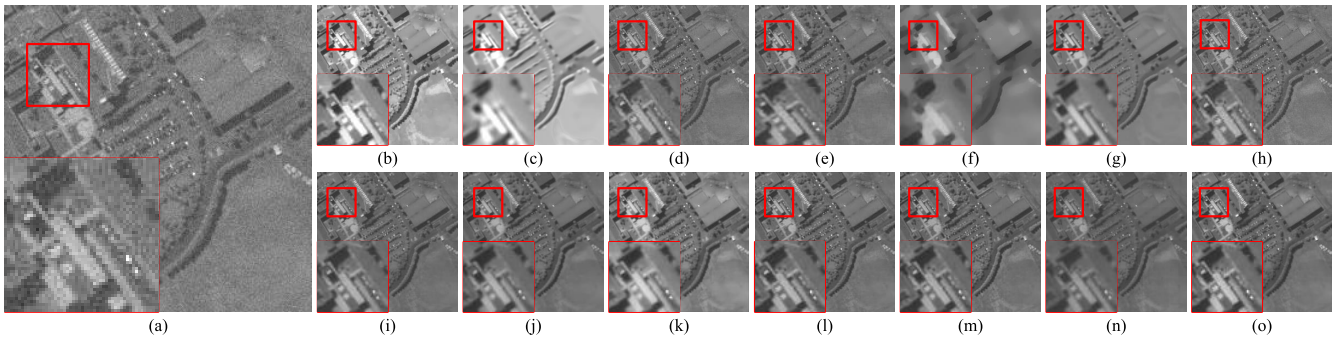


Fig. 12. Denoising results for Pavia University dataset with mixed noise, grayscale display with band 73. (a) Noisy image. (b) E3DTV. (c) LLRT. (d) LRMR. (e) LRTDTV. (f) LRTV. (g) NGmeet. (h) NMoG. (i) QRNN3D. (j) TRQ3D. (k) SST. (l) MAC-Net. (m) NSSNN. (n) SERT. (o) S^2OTAN.

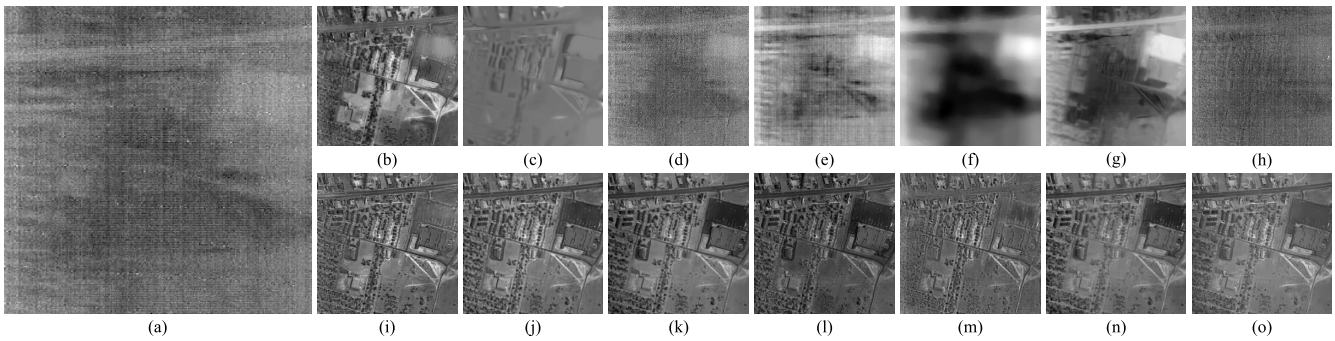


Fig. 13. Denoising results for Urban dataset in the real-data experiment, grayscale display with band 105. (a) Original image. (b) E3DTV. (c) LLRT. (d) LRMR. (e) LRTDTV. (f) LRTV. (g) NGmeet. (h) NMoG. (i) QRNN3D. (j) TRQ3D. (k) SST. (l) MAC-Net. (m) NSSNN. (n) SERT. (o) S^2OTAN.

to oversmoothing in the results of LRTV. CNN-based models, such as QRNN3D and MAC-Net, exhibit strong local focusing capabilities. However, these models still suffer from residual stripes in their denoising results. On the other hand, models like TRQ3D and SERT, with the aid of attention mechanisms, contribute to noise reduction in the regions of interest. Nonetheless, they tend to exhibit excessive smoothing effects. By contrast, SST, NSSNN, and the proposed method perform visually better by effectively removing structural noise while retaining spatial information.

E. Ablation Study

To evaluate the details in the proposed method, we further conduct ablation experiments to investigate the impacts of individual components and parameters. All experiments are performed in Case 1.3 of the CAVE dataset.

1) Impact of Modules: SW-MSA, CW-MSA, and IS-MSA are the main components of our method, which can be abbreviated as S, C, and I, respectively. The residual network that only contains model S can be regarded as the baseline model of our method. The specific quantitative analysis is shown in Table V, which indicates that introducing model C into the baseline model helps to enhance the information exchange among windows and fully utilize the nonlocal similarity in space. Model I can further extract the spectral correlations of HSIs, thereby achieving better denoising effects. Our method leverages the advantages of both models to capture interchannel relationships, gather global information, and enhance representation capability. The effective synergistic interaction among the components is intuitively reflected in the improvement of the indicators.

2) Impact of Modules Arrangement Structures: Currently, many works employ a combination of spatial attention and spectral attention [41], [42]. However, the superiority of parallel or sequential structures for these two types of attention is worth discussing. Therefore, ablation experiments are con-

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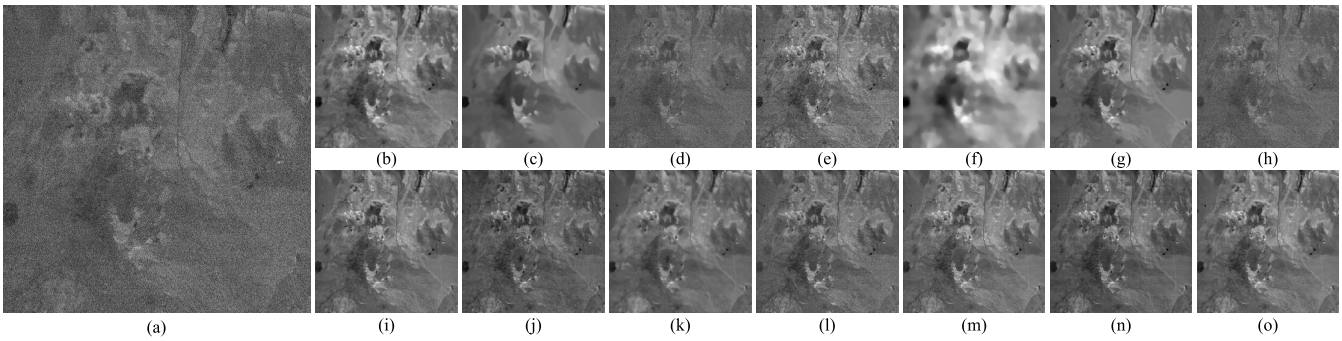


Fig. 14. Denoising results for Cuprite dataset in the real-data experiment, grayscale display with band 223. (a) Original image. (b) E3DTV. (c) LLRT. (d) LRMR. (e) LRTDTV. (f) LRTV. (g) NGmeet. (h) NMoG. (i) QRNN3D. (j) TRQ3D. (k) SST. (l) MAC-Net. (m) NSSNN. (n) SERT. (o) S^{2}OTAN.

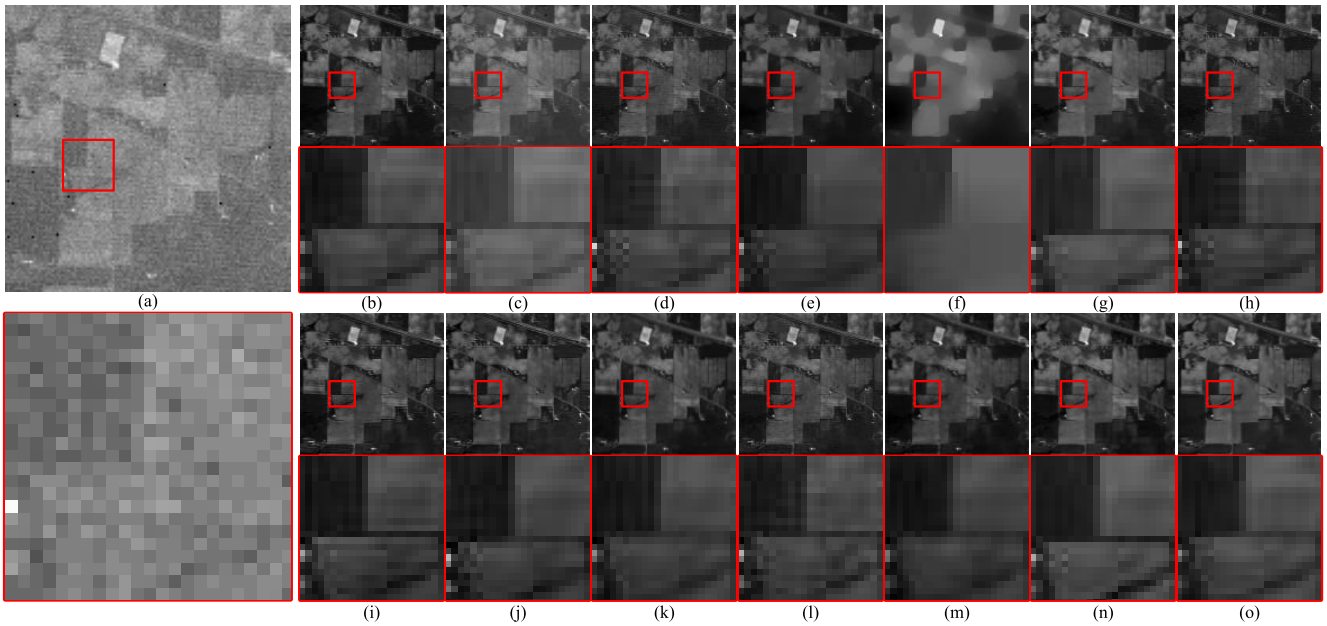


Fig. 15. Denoising results for Indian Pines dataset in the real-data experiment, grayscale display with band 3. (a) Original image. (b) E3DTV. (c) LLRT. (d) LRMR. (e) LRTDTV. (f) LRTV. (g) NGmeet. (h) NMoG. (i) QRNN3D. (j) TRQ3D. (k) SST. (l) MAC-Net. (m) NSSNN. (n) SERT. (o) S^{2}OTAN.

TABLE V

QUANTITATIVE EVALUATION RESULTS FOR MODULES ABLATION STUDY IN CASE 1.3

Model	PSNR	SSIM	SAM	ERGAS
Baseline	39.35	0.9947	0.1789	54.40
Baseline+C	40.08	0.9983	0.1672	52.44
Baseline+I	39.89	0.9982	0.1748	53.46
Baseline+C+I	**40.30**	**0.9984**	**0.1602**	**50.83**

TABLE VI

QUANTITATIVE EVALUATION RESULTS OF DIFFERENT ARRANGEMENT STRUCTURES OF COMPONENTS IN CASE 1.3

Structure		PSNR	SSIM	SAM	ERGAS
In sequential	(S+C)-I	40.18	0.9983	**0.1527**	51.42
	I-(S+C)	39.38	0.9980	0.1787	56.36
In parallel	(S+C)-I	**40.30**	**0.9984**	0.1602	**50.83**

ducted on the module arrangement structure, as shown in Table VI. In the experiments, models S and C are included in one branch, while model I is included in another branch. It can be observed that, regardless of whether model I is placed before or after, the sequential structure performs slightly inferior than the parallel structure. This finding provides evidence for the effectiveness of the parallel structure we adopted.

3) Impact of the Network Depth: With the increasing network depth in the transformer block, the number of parameters and floating points increases accordingly. The capacity of the network to extract features will also vary. Therefore, choosing an appropriate depth can maximize performance while saving computational costs. For this purpose, we conduct experiments with different depths, setting 3, 4, 6, and 8 layers in the proposed method for comparison. Table VII displays the comparison results in image quality and parameter quantity. It is seen that the model with six layers accesses the optimal results, which serves as the final criterion for the proposed method.

4) Sensitivity Analysis of Embedding Dimension C: Accurately estimating the value of the embedded dimension C is a challenging task. An overestimated C may cause incomplete utilization of the LR property of the spectrum, which decreases the denoising performance. Conversely, an underestimation of C often brings about increased computational complexity

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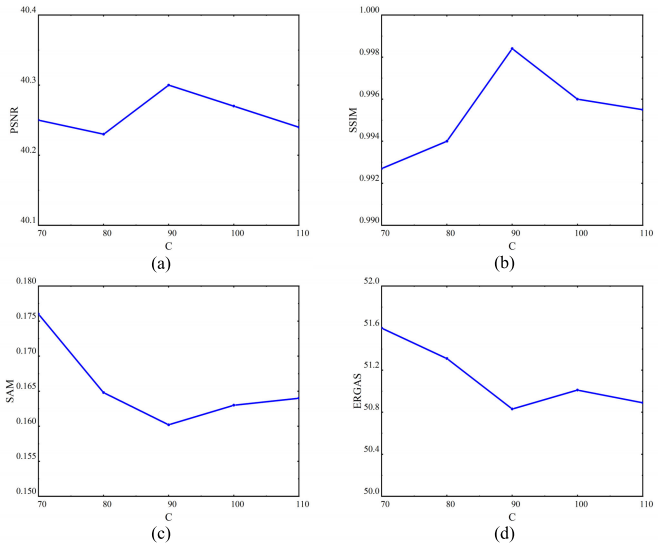


Fig. 16. Sensitivity analysis of parameter C in case 1.3. (a) PSNR. (b) SSIM. (c) SAM. (d) ERGAS.

TABLE VII
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT NUMBER OF LAYERS IN CASE 1.3

Layers	PSNR	SSIM	SAM	ERGAS	Param (M)	GFLOPs
3	37.71	0.9927	0.1960	93.81	1.83	10.10
4	38.65	0.9940	0.1648	83.65	2.25	12.73
6	**40.30**	**0.9984**	**0.1602**	**50.83**	3.09	17.91
8	39.91	0.9982	0.1618	53.21	3.80	22.52

and more significant noise interference, also leading to subpar performance. Building upon previous works [41], we selected a range of values for experimental analysis, as shown in Fig. 16, and eventually chose 90 as the final value for the embedded dimension C.

V. CONCLUSION

In this work, we introduce a novel method for removing mixed noise from HSIs, called S^{2}OTAN. Compared with other transformer-based approaches, the proposed method focuses on the prior knowledge of HSIs and uses spatial–spectral constraints to remove mixed noise robustly. Specifically, our network is built upon a residual architecture, which provides deeper feature information for image restoration and ensures the stability of network training. Then, SSTAB is used to extract HSI features from both spatial and spectral dimensions. In terms of spatial dimension, the SW-MSA and CW-MSA layers are used to compensate for the partial information extraction ability of the transformer and achieve cross-window information interaction without introducing additional computational complexity. Regarding the spectral dimension, we use the IS-MSA layer to divide the spectrum and further enhance local spectral correlation. Experimentally, a comparison of simulation and real-mixed noise scenarios demonstrates that the proposed method performs robust denoising in multiple scenarios, particularly in detail preservation and spectral fidelity, outperforming competing methods. Given the powerful feature extraction capabilities of the transformer for learning prior, in the future, we will extend the proposed approach to higher level remote sensing missions, for instance, image classification and object detection.

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